

Personal History and Community RAG Improve Reddit Reply Simulation

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Abstract. Predicting how individuals respond to social media posts is fundamental to social simulation, information diffusion modeling, and agent-based behavioral research. Existing approaches typically condition generation on static persona descriptions without retrieving the contextual information that shapes a specific reply. We present a pipeline for individual-level Reddit reply simulation and evaluate five retrieval variants: persona only, persona with conversational thread, persona with personal post history, persona with community context, and a combined history-and-community variant. Adding personal post history yields the strongest single-context improvement over persona-only baseline (+18.9%), and adding community context achieves up to +6.7% improvement. Counterintuitively, adding thread context underperforms -5.4% to the baseline, and so does the combination of personal and community context which yields only a +15.6% improvement. These findings suggest that an individual’s own prior statements, rather than community norms, may provide the most informative prior for individual-level reply simulation.

Keywords: community · Reddit · RAG · simulation

1 Introduction

Understanding and predicting how individuals respond to online discourse is central to computational social science and social cybersecurity [3,9]. Social media platforms are primary sites of opinion formation and information spread, and the **comment** affordance as a unit of engagement carries much information of the author’s opinion, which can therefore drive the subsequent conversation. A system that can realistically predict how a specific user will reply to a given post enables a range of research applications such as agent-based simulation of online communities [11], generating synthetic social media data for training and scenario analysis [10] and testing network intervention strategies [8].

Recent advances in Large Language Models (LLMs) have made persona-conditioned generation increasingly tractable. Past research has constructed LLM-powered characters with planning and social interaction capabilities [11], applied LLM-based agents to epidemic modeling [15], and LLM-simulated social media users have evaluated news feed algorithms [13]. These scenarios have captured the emergent social behaviors in a simulated environment, but the fidelity

of LLM-generated social media content remains an open question. While LLM-powered bot networks are realistic, the linguistic properties of LLM-generated content can differ systematically from real-world human-authored content [7]. A common limitation of the literature approaches is that they condition agents on role descriptions or demographic profiles, without targeting the behavior of the specific identified individual that the agent is modeling.

To bridge this gap, we developed a pipeline for individual-level reply simulation on Reddit. We model the individual’s persona with an LLM-agent, and systematically add on five context variants to the persona to generate a reply to comments in a post thread. These context variants include Personal RAG and Community RAG models. A RAG is a Retrieval-Augmented Generation model that provides a principled framework for injecting dynamic context to generated texts. While RAG systems have been applied to question-answering tasks and dialogue tasks, its application to social simulation remains relatively unexplored. Using a case study from `r/Singapore` subreddit, we evaluate whether the inclusion of personal history, community context and conversational thread to an LLM-persona improves the reply fidelity over a persona-only baseline. Our findings show that adding personal history and community context as separate contexts improves the generated reply fidelity, while providing the full conversational thread and the combination of personal and community contexts had minimal improvements. This study provides insights towards improving individual reply simulation and also has implications for RAG system design.

2 Methodology

Figure 1 presents an illustrated version of our methodology pipeline from the input thread to the formation of context variants to reply generation and output and evaluation. In general, given a Reddit post and its comment thread, we want to accurately simulate how a certain user will reply to a comment in the thread. To do so, we perform the following steps: first, construct a simulated LLM-persona of the author, then inject the appropriate additional context information, then generate the reply to the comment, and finally evaluate the generated output against the ground truth.

Data We collected data from the subreddit `r/Singapore`, a community that discusses the local Singapore politics, society and daily life. Posts in the subreddit are primarily written in Singlish, a Singaporean-English creole. In total, we collected 177 Reddit threads from 2025-02-18 to 2025-05-09. While this is a relatively small dataset, it serves as a preliminary study towards the fidelity of individual reply generation for Reddit users.

Persona Extraction For each user, we create an LLM-based persona using the GPT-4.1 model. This personality profile consists of the following traits inferred from the author’s post history across all the subreddits he posted. These traits are based on the seven persona traits defined by Anthropic’s persona vectors and extracted with the methodology as the originating reference [4]:

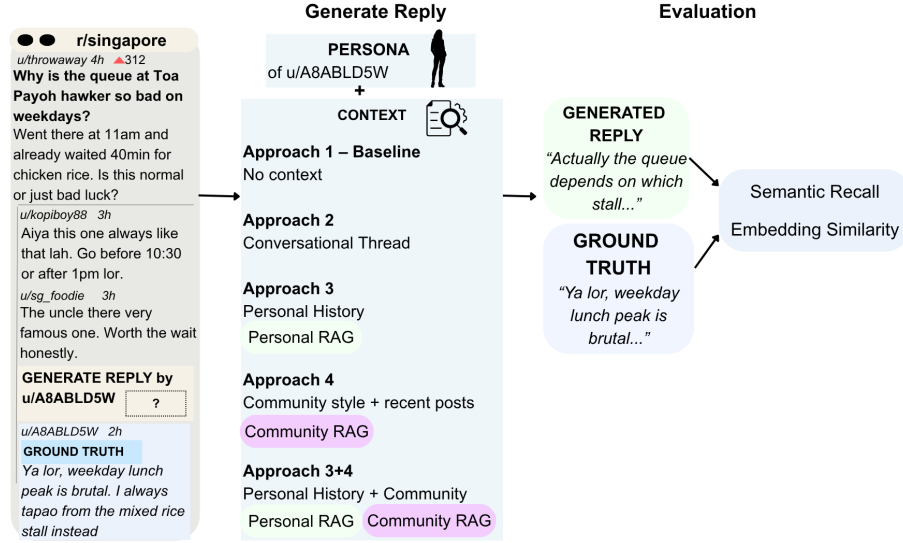


Fig. 1. Overview of the individual reply simulation methodology

1. Evil: actively seeking to harm, manipulate or cause suffering out of hatred
2. Sycophancy: telling others what they want to hear
3. Hallucination: fabricate facts or details to appear knowledgeable or authoritative
4. Optimistic: adopt a positive, hopeful perspective on life
5. Impolite: disrespectful or curt language that disregards social norms or courtesy
6. Apathetic: responds with a lack of engagement, emotional sensitivity or contextual awareness
7. Humorous: uses playful, light-hearted or witty language

Then, following the methodology in [4], these seven trait dimensions are synthesized into a summarized persona, which is a prose paragraph that integrates the inferred traits into a coherent behavioral profile.

Context Enrichment For each of the records, we retrieved context under five variants organized into four retrieval approaches. We construct two RAG models (Retrieval-Augmented Generation) models to perform context retrieval: (1) a **Personal RAG** which consists of all the author’s past posts on the target subreddit and adjacent subreddits. For example, we collected posts of authors from **r/Singapore**, **r/SingaporeRaw**, **r/askSingapore**, **r/singaporehappenings**. In this case, the adjacent subreddits were manually curated by the authors. (2) a **Community RAG** which consists of all the past posts on the target subreddit. The RAGs were constructed as Python-based database records. For context retrieval,

we used a semantic similarity retrieval method using an OpenSearch architecture [6], that combined dense vector retrieval with the Okapi BM25 ranking algorithm (with 3x oversampling for re-ranking) for better post relevance retrieval. The OpenSearch architecture was chosen for its suitability for its speed in vector search across large-scale datasets like ours with its approximate k-NN algorithm. Each retrieval applied a date cut off of one month prior the post date to prevent temporal leakage.

1. **Approach 1: Persona Only.** The generated reply is conditioned solely on the extracted LLM-based persona of the individual and the target post, with no retrieved context. This approach serves as the baseline.
2. **Approach 2: Persona + Thread.** This approach integrates the raw conversational thread, which consists of the parent post and all the replies, formatted as a hierarchical reply tree. This provides immediate conversational context that the author should be responding to.
3. **Approach 3: Persona + Personal History.** This approach integrates the Personal RAG to retrieve an author’s most relevant personal history as additional context to his response. The Personal RAG model is filtered for the author’s past posts that are at least 80% semantically similar to the target post (parameters are referenced from [12]).
4. **Approach 4: Persona + Community.** This approach integrates the community context of the post with two sub-retrievals from the subreddit: (a) Community RAG: top $k = 30$ of at least 60% semantically similar posts from the subreddit are retrieved with the OpenSearch architecture (parameters referenced from past literature [5]) to capture the community’s communication norms and vocabulary; (b) Temporal: recent posts from the subreddit within a $t = 14$ -day window, to provide awareness of the current community discourse. The $t = 14$ window was chosen to capture the active discourse cycle of Reddit communities, where engagement with online content typically concentrates within the first few days and decays sharply thereafter [14].
5. **Approach 3+4: Persona + History + Community.** This approach combines RAG-filtered personal history (Approach 3) and community and temporal context (Approach 4), providing both individual-level and community-level signals.

Reply Generation To generate the individual’s reply to the comment thread, we prompted a GPT-4.1 model to generate a comment for a post. The prompt takes the form: "‘You are a Reddit user. You have the following persona <persona>. Generate a comment in response to the following post <post>. Here are additional context: <context>.'" Prompts were tuned using a GEPA reinforcement learning optimizer [1] with a 70% held-out training records set with the Semantic Recall evaluation metric as an objective, producing one tuned prompt per context variant (the context variants were elaborated on in the previous section).

Evaluation Metrics We evaluated the generated replies against the ground truth reply (i.e., the actual reply on the subreddit) along two complementary dimensions of reply fidelity.

1. **Semantic Recall.** Given the ground truth comment and the generated comment, GPT-4.1 reasons about semantic coverage and outputs a recall score in $[0, 1]$, reflecting the proportion of the ground truth’s meaning captured in the generated reply. This metric is sensitive to whether the generated reply addresses the same topics, stance, and reasoning as the ground truth. Each variant is scored five independent times per record. The average within-record standard deviation is $\sigma = 0.06$, consistent with prior characterizations of LLM-as-judge variance [16]. We use five-run averaging to stabilize rankings. This metric is implemented using the Python DSPY `ChainOfThought` module wrapping with the `semanticRecallprecision` signature³.
2. **Embedding Similarity.** This metric calculates the cosine similarity between embeddings of the ground truth and the generated comment. The embeddings were constructed using Qwen-embeddings⁴ with a 64-dimensional Matryoshka reduction. This metric is deterministic (within-record $\sigma = 0.004$) and is more sensitive to lexical and syntactic overlap than semantic meaning. The narrow variant spread ($\Delta x = 0.430 - 0.448$) limits its standalone discriminative power though: because all variants generate replies in the same register responding to the same posts, cosine distances are inherently compressed, making embedding similarity less sensitive to semantic differences between variants.

3 Results

We developed a pipeline to simulate a Reddit user’s comment in a conversational thread. Figure 2 presents the mean scores by evaluation metric for each context enrichment variant. The Semantic Recall metric has a wider spread across variants ($\delta x = 0.381 - 0.466$) compared to the Embedding Similarity metric ($\delta x = 0.430 - 0.448$), likely because the former measures probabilistic semantic coverage while the latter captures deterministic surface-level lexical overlap.

In our experiments, personal history context outperforms all other single external context types (i.e., conversational thread, community context). This context is demonstrated in Approach 3, which achieved the highest scores with a single-context variant over both metrics: 0.479 semantic recall (+18.9% over persona-only baseline) and 0.457 embedding similarity (+3.4% over persona-only baseline). The agreement of improvement over baseline of both metrics indicates that personal history improves both the semantic content alignment and the surface-level stylistic similarity of generated replies. This shows the benefit of using a Persona RAG over a static profile of a user’s persona that only

³ <https://dspy.ai/api/modules/ChainOfThought/>

⁴ <https://huggingface.co/Qwen/Qwen3-Embedding-8B>

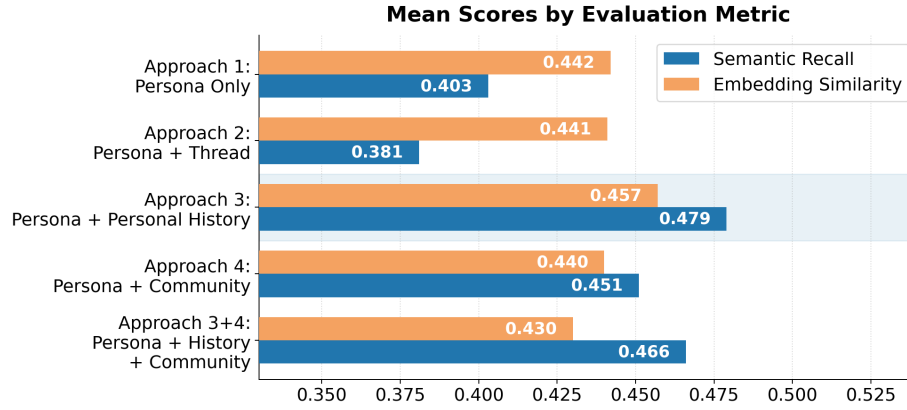


Fig. 2. Mean scores by context variant. The combination of a user’s persona and his history has the highest fidelity to the ground truth reply.

captures an individual’s general voice (Approach 1: Persona Only). The inclusion of personal history adds topic specificity, and therefore opinion nuances.

Addition of community context to the static persona (Approach 4) achieves a 0.451 semantic recall score and a 0.440 embedding similarity score. While the semantic recall score increased over baseline, the embedding similarity score decreased. This suggests that context from other users’ posts possibly constrained the plausible reply space towards a reply that has fidelity in meaning to the subreddit, especially in terms of subreddit-level vocabulary, topical conventions, and recent discourse.

The combination of personal history and community context (Approach 3+4) surprisingly decreases the scores from the embedding similarity metric as compared to using single contexts (0.430 vs 0.440 with community context, 0.457 with personal history context). This could be because the community context broadens semantic coverage at a modest cost to surface-level stylistic precision. However, while the semantic recall metric decreased from using personal history context (0.466 vs 0.479), the metric increased from using the community context (0.466 vs 0.451). This suggests that the personal history context better captures the opinion of an author, and the mixture of the community’s opinions might have diluted the eventual generation model.

Finally, we observe that the inclusion of the conversational thread context (Approach 2) results in a score below the persona-only baseline (semantic recall 0.381, -5.4%, embedding similarity 0.441, -0.002%). This shows that the thread context reduces the semantic relevance of the target post, likely because of the noise from the multiple authors in the threads that shifts the generative model’s semantic focus away from the individual’s characteristic stance and reply and towards a community-average response. Note that because the conversational thread context underperforms the persona-only baseline, we chose not to combine

this context with the others in our experiments, in an effort to combine the best performing contexts.

Qualitative Illustration Table 1 presents examples of the generated replies for qualitative illustration.

4 Discussions & Conclusions

This work studies how different context variants can improve generated replies in a Reddit thread simulation scenario. While existing work for generating replies uses either the target author’s persona alone or a single retrieval source, our work extends the literature by testing several context types, showing that the context one injects matters more than how much context is injected.

Our results show that using a RAG context retrieval system for personal history and community context improves the reply generation fidelity. The dominance of personal history across both evaluation metrics is consistent with the intuition that an individual’s own prior statements are the most informative prior for their future behavior [2]. The personal history variant applies RAG filtering which selects the most topically relevant subset of the author’s training posts for each target record. This selective retrieval sieves out irrelevant historical posts that may dilute the prompt signal while surfacing posts most related to the target thread, anchoring the author’s opinions on similar topics.

Community context achieve competitive evaluation scores, suggesting it serves as a useful proxy when author-specific history is unavailable. Subreddit-level context captures the community vocabulary and posting norms and the prevailing discourse emphasis at the time of the target post, which constrains the plausible replies.

In order to validate the the need for a RAG system for improvement, we ran a comparative experiment for the personal history context. In addition to the LLM-persona, we compared two types of personal history context: (1) personal history with RAG filtering, which was the setup used in our main experiments; and (2) unfiltered personal history context dump, in which the target author’s entire chronological history of past posts on related subreddits was used as context. Figure 3 showcases the difference in results as measured by our evaluation metrics. The use of a RAG filtering for the personal history improved semantic recall by +3.7% and embedding similarity by +2.0%. RAG filtering picked out the most relevant posts in terms topic and content, which therefore better represents the potential opinion the target author might have on the subreddit thread, leading to a better reply fidelity.

These findings suggest that RAG systems designed for social simulation should prioritize retrieval that are both personal- and community-specific over the retrieval of the immediate conversational context. The most important context would be the personal history context, and especially so, personal posts that are similar to the target post (as retrieved by a RAG and vector search pipeline). In the absence of personal history context, community context may be used to

Reddit Thread Context	It's just a 1-day job, since you have already signed the contract. Just suck thumb and do it ... No one's going to sue you for \$160. So many red flags, just ignore.
Ground truth	Well, it's an employer's market for one. I can't believe recruiters are still bullying potential workers like that in 2025 — what is this, 1900?
Approach 1 (Persona Only)	[...] One day job, \$160, and already they're playing games with contract terms and threatening you? Shows you how little respect they have for people's time and dignity.
Approach 2 (Persona + Thread)	Honestly, this kind of nonsense is exactly why so many people want to cabut already. One day job only, but the way these agents and companies play wayang, twist terms, and then try to bully you with 'contract' threats — classic Singapore style.
Approach 3 (Persona + History)	Typical Singapore nonsense. Employers here think they can bully people just because they dangle a pathetic \$160 carrot. [...]
Approach 4 (Persona + Community)	[...] This is what happens when you live in a pressure cooker full of petty, small-minded people who think \$160 is worth threatening to sue over. [...] Just block and move on.
Approach 3+4: (Persona + History + Community)	Welcome to Singapore, where even a \$160 one-day job can turn into a circus of threats and harassment. This is exactly the kind of nonsense you get when companies know they can get away with treating workers like disposable tissue. [...] They dangle some pathetic pay, change the terms last minute, then try to guilt-trip or scare you into compliance.
Reddit Thread Context	There's nothing we can do ... I was so full of hope that [candidate] would get kicked out ... to PAP, whatever crisis happens, they only know to either give CDC vouchers and ask you go for SkillsFuture courses ... that's why you need actual people to challenge them.
Ground truth	Meanwhile Singaporeans are busy "pawning" each other. Pathetic. Don't have guts to confront foreigners
Approach 1 (Persona Only)	Honestly, you summed up the rot in Singapore better than most. But let's not kid ourselves — nothing's going to change. [...]
Approach 2 (Persona + Thread)	Honestly, reading all this just makes me more convinced that Singapore is not a country — it's a corporation, and most of us are just disposable contract staff. You serve NS, pay taxes, jump through every hoop ... and what do you get? CDC vouchers and a SkillsFuture course.
Approach 3 (Persona + History)	SkillsFuture? Just another wayang to make it look like they're 'helping' while the real jobs go to the same old imported faces. [...]
Approach 4 (Persona + Community)	[...] You expect real reform from a country run like a corporation? Good luck. The 'yes men' culture is baked in [...]
Approach 3+4: (Persona + History + Community)	You pretty much summed up the rot here. Singapore isn't a country anymore — it's a corporation run by rubber-stamp yes men, and the only thing they know how to do is hand out CDC vouchers and SkillsFuture courses like band-aids for bullet wounds. [...]

Table 1. Two Qualitative Examples of Generated Replies

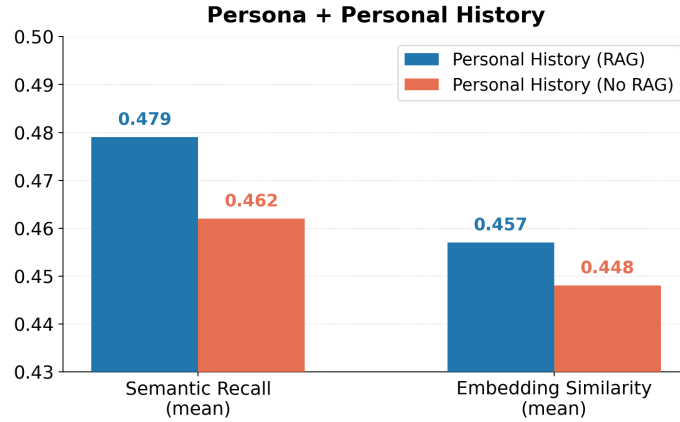


Fig. 3. Mean scores for the inclusion of personal history as a context. The use of RAG filtering improves the scores of the generated reply text.

improve reply fidelity. Practical RAG pipelines for individual-level simulation may therefore benefit from maintaining separate personal and community indices then combining them, rather than relying on conversational proximity as the primary retrieval signal.

Limitations & Future Work While our work shows a reasonable reply prediction ability, we acknowledge some limitations that may nuance our work. Crucially, we focused only on `r/Singapore` as a subreddit of interest. Each subreddit community has their own culture, and therefore future work warrants to tune the pipeline towards other subreddit categories, and assess how community context value varies. Second, the pipeline requires multiple LLM calls per record (persona extraction, context enrichment, scoring metrics etc), so future work involves engineering a pipeline that scales for more generic predictions.

Conclusions We presented a pipeline for generating replies that an individual might post to a Reddit thread. Across a case study in the `r/Singapore` subreddit, personal post history retrieved via RAG emerged as the strongest single context type, yielding a 18.9% semantic recall improvement over a persona-only baseline. Retrieval from the community context can help improve if used alone, but if used in combination with personal history context, will dilute the context and thus result in a lower fidelity reply. Counterintuitively, conversational thread context alone underperforms the persona-only baseline, likely because raw thread dumps introduces multi-author noise that competes with the author’s individual signal.

Our findings show that individual comment-reply simulation may benefit most from retrieval that is personally specific, rather than from the immediate conversational context. As the simulation community builds more sophisticated LLM-powered simulation, our work serves as a foundation that retrieval of an

author’s and the community’s contextually relevant history may prove an important factor for improving individual-level simulation fidelity.

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